OYM NBTS TEST-8 (Rolex)

1. If the block start sliding on an rough inclined surface when angle of inclination becomes θ, then value of coefficient of static friction will be

a. cot(θ)
b. tan(θ)
c. cos(θ)
d. sin(θ)

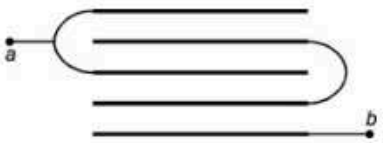
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2. A magnetising field intensity of 10 A m^{-1} is applied on a sample of ferromagnetic material of susceptibility 1999. The net magnetic field inside the material is

a. $4\pi \times 10^{-6}$ T
b. $8\pi \times 1999 \times 10^{-6}$ T
c. $8\pi \times 10^{-3}$ T
d. $\pi \times 1998 \times 10^{-6}$ T

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3. Five plates each of area A are arranged as in the diagram. If the spacing between any two consecutive plates is d, then equivalent capacitance between a and b is



a. $\frac{3A\varepsilon_0}{2d}$
b. $\frac{3A\varepsilon_0}{4d}$
c. $\frac{2A\varepsilon_0}{3d}$
d. $\frac{4A\varepsilon_0}{3d}$

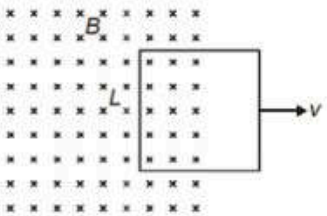
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4. Potential in the y-z plane is given as $V = 5\ (y + z)$ where V is in volt and y, z are in meter. The magnitude of electric field at (0, 1 m, 1 m) is

a. 5 V/m
b. $5\sqrt{3}$ V/m
c. 10 V/m
d. $5\sqrt{2}$ V/m

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5. A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A constant magnetic field B exists as shown in the figure. Mechanical power required to maintain its uniform velocity is



a. $\frac{B^2L^2v^2}{R}$
b. $\frac{B^2L^2v^2}{2R}$
c. $\frac{B^2L^2v^2}{4R}$
d. Zero

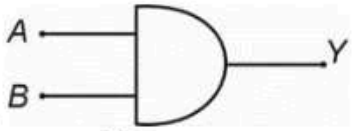
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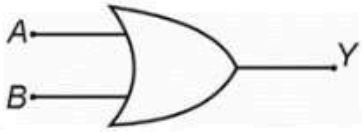
6. Lorentz force is given by

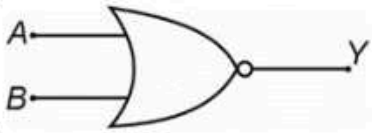
a. $\vec{F} = q[2\vec{E} + (\vec{B} \times \vec{v})]$
b. $\vec{F} = q[\vec{E} - (\vec{v} \times \vec{B})]$
c. $\vec{F} = q[\vec{E} - (\vec{v} \cdot \vec{B})]$
d. $\vec{F} = q[\vec{E} + (\vec{v} \times \vec{B})]$

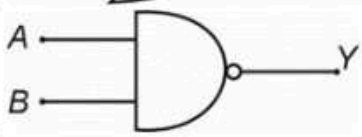
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7. Which is correct symbol of NAND gate?

a. 

b. 

c. 

d. 

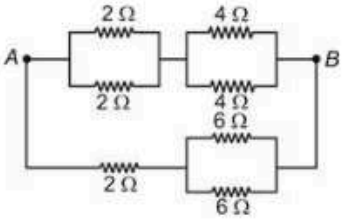
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8. Charge through a cross-section of a conductor is given by $Q = (2t^2 + 4t)$ coulomb. The average current through the conductor from $t = 0$ to $t = 3$ second is

a. 10 A
b. 30 A
c. $\frac{16}{3}$A
d. Zero

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9. The equivalent resistance across AB is



a. 15 Ω
b. 8/15 Ω
c. 1/15 Ω
d. 15/8 Ω

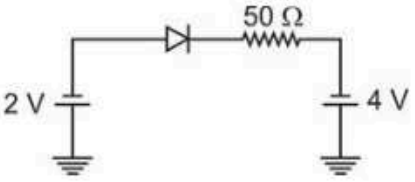
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10. The de Broglie wavelength of a particle moving with a velocity of 2×10^8 ms^{-1} is equal to the wave length of photon. What will be ratio of kinetic energy of the moving particle to that of energy of a photon?

a. 1
b. $\frac{1}{4}$
c. $\frac{1}{3}$
d. $\frac{2}{3}$

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11. What is the current through an ideal $P-N$ junction diode shown in figure below?



a. 40 mA
b. 20 mA
c. 10 mA
d. 80 mA

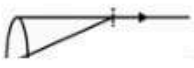
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12. If the nuclear radius of ^{27}Al is 3.6 fermi, then the approximate nuclear radius of ^{64}Cu (in Å) is

a. 4.8
b. 4.8×10^{-5}
c. 4.8×10^{-3}
d. 4.8×10^{-4}

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13. A cylindrical solid rod made up of material of refractive index $\frac{2}{\sqrt{3}}$ is given. A light beam enters from one circular end of the cylinder, and comes out just brushing the outer surface along the length. What should be the angle of incidence on the circular face?



14. Waves from two coherent sources with intensity $4I_0$ and $9I_0$ respectively, interfere at a point in medium. The minimum intensity would be

a. $5I_0$
b. $4I_0$
c. $2I_0$



a. $\sin^{-1}(\frac{1}{2})$
b. $\sin^{-1}(\frac{1}{\sqrt{3}})$
c. $\sin^{-1}(\frac{\sqrt{3}}{2})$
d. $\sin^{-1}(\frac{2}{\sqrt{3}})$

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d. I_0

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15. If ionisation energy of a hydrogen atom is 13.6 eV. What is the energy of a level with quantum number n = 3?

a. 0.85 eV
b. – 1.51 eV
c. – 10.2 eV
d. – 24.8 eV

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16. If an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit, it emits a photon of wavelength λ. When it jumps from the 4th orbit to the 3rd orbit, the corresponding wavelength of the photon will be

a. $\frac{16}{25}\lambda$
b. $\frac{9}{16}\lambda$
c. $\frac{20}{7}\lambda$
d. $\frac{20}{13}\lambda$

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17. For a uniformly charged ring of diameter R, the electric field on its axis has the maximum magnitude at a distance x along axis of ring from its centre. The value of x is

a. $\frac{R}{2\sqrt{2}}$
b. $2\sqrt{2}R$
c. $\frac{R}{\sqrt{2}}$
d. $\sqrt{2}R$

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18. The velocity V is given by $V = at^2 + bt + c$, where t is time. The dimension of $\frac{b}{c}$ is

a. $[M^0L^0T^1]$
b. $[M^0L^0T^{-1}]$
c. $[M^0L^1T^0]$
d. $[M^0L^{-1}T^0]$

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19. The distance travelled by a particle starting from rest and moving with an acceleration $\frac{4}{3}\ ms^{-2}$ in the first second is

a. 6 m
b. $\frac{10}{3}$ m
c. $\frac{2}{3}$ m
d. $\frac{19}{3}$ m

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20. A body is projected with a velocity of 60 ms^{-1} at an angle 30° with horizontal. Based upon above information some quantities are given in column-I and their vector notation in the column-II. Match the correct entries in column-I with column-II.

Column-I	**Column-II**
(i) Initial velocity vector	(a) $60\sqrt{3}\hat{i} + 40\hat{j}$
(ii) Velocity after 3s	(b) $30\sqrt{3}\hat{i} + 10\hat{j}$
(iii) Displacement after 2s	(c) $30\sqrt{3}\hat{i} + 30\hat{j}$
(iv) Velocity after 2s	(d) $30\sqrt{3}\hat{i}$

a. i – a, ii – c, iii – d, iv – b
b. i – c, ii – a, iii – d, iv – b
c. i – c, ii – d, iii – a, iv – b
d. i – c, ii – d, iii – b, iv – a

a. I – c, II – d, III – b, IV – a

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21. A man who can swim at the rate of 2 km/h (in still water) wishes to cross a river to a point exactly opposite on other bank by swimming. If the velocity of water current is 1 km/h, then the direction in which he should swim is

a. 30° with the flow of river water
b. 60° with the flow of river water
c. 120° with the flow of river water
d. 90° with the flow of river water

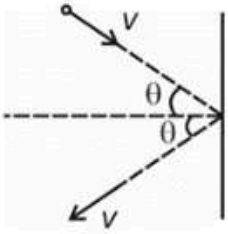
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22. The angle made by vector $\hat{i} + \sqrt{3}\hat{j}$ with the x-axis is

a. 45º
b. 30°
c. 60°
d. 37°

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23. A particle of mass m moving with a velocity v strikes a wall at an angle of incidence θ and rebounds back as shown. The magnitude of impulse imparted to the particle by the wall is



a. mv cosθ
b. 2 mv cosθ
c. mv sinθ
d. 2 mv sinθ

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24. If U represents the potential energy associated with a particle constrained to move under the action of conservative force, F, then in equilibrium

a. $\frac{dU}{dr} \neq 0$
b. $\frac{dU}{dr} > 0$
c. $\frac{dU}{dr} = 0$
d. $\frac{dU}{dr} < 0$

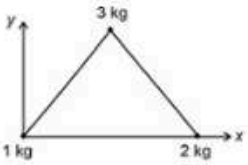
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25. A particle of mass 5 kg is moving with uniform speed $3\sqrt{2}$ m/s in xy plane along the line $y = x$. The magnitude of its angular momentum about origin is

a. 40 units
b. 60 units
c. Zero
d. $40\sqrt{2}$ units

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26. Three particles of masses 1 kg, 2 kg and 3 kg are situated at the corner of an equilateral triangle of side b as shown in the figure. The y-coordinate of the centre of mass is



a. Zero
b. $\frac{3b}{6}$
c. $\frac{3\sqrt{3}b}{12}$
d. $\frac{3\sqrt{3}b}{5}$

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27. A satellite of mass m is revolving around the earth of mass M in a circular orbit of radius $2R$. The amount

28. Two wires A and B have same length and same area of cross-section. Young's modulus of A is two

of energy given to the satellite so that it starts revolving in circular orbit of radius $4R$ is

a. $\frac{GMm}{4R}$
b. $\frac{GMm}{8R}$
c. $\frac{GMm}{2R}$
d. $\frac{GMm}{6R}$

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times the Young's modulus of B. If they are subjected to same external load, then the ratio of stress produced in A to that in B is

a. 2 : 1
b. 3 : 5
c. 1 : 1
d. 3 : 7

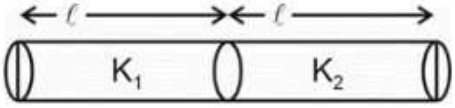
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29. The coefficient of volume expansion of material of a metre scale is γ. The fractional change in its length when temperature is changed by ΔT is

a. $\gamma\Delta T$
b. $\frac{\gamma}{2}\Delta T$
c. $\frac{\gamma}{3}\Delta T$
d. $3\gamma\Delta T$

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30. Consider a compound slab consisting of two different material having equal area of cross-section and thermal conductivities being K and $4K$. The equivalent thermal conductivity of slab is



a. $\frac{K_1 + K_2}{2}$
b. $\frac{2K_1K_2}{K_1 + K_2}$
c. $\frac{K_1 + K_2}{2K_1K_2}$
d. $\frac{K_1K_2}{K_1 + K_2}$

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31. Choose the correct statement among the following

a. All heat cannot be converted into work
b. The efficiency of heat engine is always greater than unity
c. The efficiency of heat engine may be greater than one
d. It is possible to transfer heat from lower to higher temperature without any work input

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32. The ratio of average translational kinetic energy of O_2 molecule at 300K to average translational kinetic energy of He molecule at 300K is

a. 1 : 1
b. 2 : 1
c. 3 : 5
d. 7 : 5

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33. The initial phase angle of a particle executing SHM according to equation $x = A\sin(\omega t + \phi)$, when it is at $+A$ and moving towards equilibrium is, (here A is displacement amplitude)

a. $\frac{5\pi}{6}$
b. $\frac{2\pi}{3}$
c. $\frac{\pi}{2}$
d. $\frac{\pi}{6}$

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34. The equation of a wave is given by $y = 10\sin(50t + 5x)$ (where x, y are in m and t is in s). The speed of travelling wave is

a. 50 ms^{-1}
b. 5 ms^{-1}
c. 10 ms^{-1}
d. 20 ms^{-1}

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35. The differential equation representing the SHM of particle is $\frac{16d^2y}{dt^2} + 9y = 0$. The angular frequency of resulting SHM is

a. 0.75 rad/s
b. 0.56 rad/s
c. 0.82 rad/s
d. 0.25 rad/s

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36. A wire of length 2 m is placed in uniform magnetic field of 2 T. It makes 30° with the field. If 10 A current is flowing in the wire, then force per unit length on wire is

a. 1 N m^{-1}
b. 0.5 N m^{-1}
c. 5 N m^{-1}
d. 10 N m^{-1}

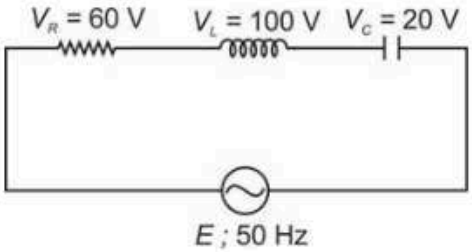
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37. A bar magnet of magnetic moment $40\hat{j}$ Am^2 is lying along *y*-axis in a magnetic field, $\vec{B} = (0.4\hat{i} + 0.3\hat{j})T$. The torque acting on the magnet is

a. $16\hat{k}$ Nm
b. $(12\hat{i} + 4\hat{j})$Nm
c. $(4\hat{i} + 3\hat{j})$ Nm
d. $-16\hat{k}$ Nm

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38. The value of alternating emf (*E*) in the given circuit will be



a. 200 V
b. 220 V
c. $\frac{220}{\sqrt{2}}$ *V*
d. 100 V

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39. The potential of the electric field at any point (*x*, *y*, *z*) is given by, $V = 3x^2 + 5$, where *x*, *y*, *z* are in metre and *V* is in volt. The intensity of the electric field at (–2, 1, 0) is

a. –12 V/m $\hat{i}$
b. +17 V/m $\hat{i}$
c. +12 V/m $\hat{i}$
d. –17 V/m $\hat{i}$

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40. If $\vec{E}$ and $\vec{B}$ represent electric and magnetic field vectors of an *EM* wave. The direction of propagation wave is along

a. $\vec{E}$
b. $\vec{B}$
c. $\vec{E} \times \vec{B}$
d. $\vec{B} \times \vec{E}$

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41. The binding energy of deuteron ^2_1H is 1.112 MeV per nucleon and an α-particle ^4_2He has a binding energy of 7.047 MeV per nucleon. Then in the fusion reaction $^2_1H + {}^2_1H \rightarrow {}^4_2He + Q$, the energy *Q* released is

a. 1 MeV
b. 11.9 MeV
c. 23.8 MeV
d. 931 MeV

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42. Sodium and copper have work functions 2.3 eV and 4.6 eV respectively. Then the ratio of their respective threshold wavelengths is nearest to

a. 1 : 2
b. 2 : 1
c. 1 : 4
d. 4 : 1

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43. The minimum magnification produced by simple microscope of focal length 5 cm is

a. 6
b. 5
c. 4
d. 10

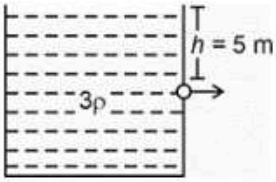
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44. In Young's experiment, fringe width was found to be 0.8 mm. If whole apparatus is immersed in water of refractive index, $\mu = \frac{4}{3}$, new fringe width is

a. 0.60 mm
b. 0.80 mm
c. 0.30 mm
d. 0.40 mm

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45. A container filled with a liquid of density 3ρ as shown. The velocity of efflux through orifice is (ρ is density of water and $g = 10\ ms^{-2}$)



a. 10 m/s
b. 15 m/s
c. 8 m/s
d. 18 m/s

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46. **Assertion (A):** The contact angle between water and glass is acute.

Reason (R): The surface of water in the glass capillary is convex.

In the light of above statements, select the **correct** option.

a. Both (A) and (R) are true and (R) is the correct explanation of (A)
b. Both (A) and (R) are true and (R) is not the correct explanation of (A)
c. (A) is true, (R) is false
d. (A) is false, (R) is true

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47. The dimensional formula for moment of force is

a. $[MLT^{-2}]$
b. $[ML^2T^{-2}]$
c. $[ML^{-2}T^2]$
d. $[ML^{-1}T^1]$

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48. A body of mass m is projected with initial speed u at an angle θ with horizontal. The magnitude of change in momentum after the body reaches its maximum height is

a. $2mu\cos\theta$
b. Zero
c. $mu\sin\theta$
d. $2mu\sin\theta$

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49. An object of mass M moving with speed u collides one dimensionally with another identical object at rest. The velocity of the object after collision which was initially at rest is (where e represents coefficient of restitution).

a. $[\frac{1-e}{2}]u$
b. $[\frac{1+e}{2}]u$
c. $(\frac{1+e}{1-e})u$
d. $(\frac{1-e}{1+e})u$

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50. The phase difference between the displacement wave and pressure wave in sound wave is

a. Zero
b. π
c. $\frac{\pi}{2}$
d. $\frac{\pi}{4}$

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51. An organic compound containing C, H and N gave the following analysis C = 40%; H = 13.33%; N = 46.67%. Its empirical formula would be

a. CH_4N
b. CH_5N
c. $C_2H_7N_2$
d. CH_3N

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52. Match the following items in column I with column II and choose the correct option.

	Column I (quantum no. s)		**Column II (orbital)**
(a)	n = 2, l = 1	(i)	3*p*
(b)	n = 3, l = 2	(ii)	3*d*
(c)	n = 1, l = 0	(iii)	2*p*
(d)	n = 3, l = 1	(iv)	1*s*

a. (a)(iii), (b)(i), (c)(iv), (d)(ii)
b. (a)(iii), (b)(ii), (c)(iv), (d)(i)
c. (a)(ii), (b)(i), (c)(iii), (d)(iv)
d. (a)(i), (b)(ii), (c)(iii), (d)(iv)

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53. In which of the following options, the order of arrangement does not agree with the variation of property indicated against it?

a. Si < P < S < Cl (increasing electronegativity)
b. Na > Li > Be > B (decreasing metallic radius)
c. Rn < Xe < Kr < Ne (Increasing electron gain enthalpy)
d. Na < Li < K < Rb (Increasing atomic radius)

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54. The heat of combustion of graphite is –440 kJ/mol. The enthalpy of formation of 15.4 g of CO_2 is

a. +154 kJ
b. –28.57
c. –154 kJ
d. +28.57

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55. Concentration of the Ag^+ ions in a saturated solution of $Ag_2C_2O_4$ is 3×10^{-4} mol L^{-1} solubility product of $Ag_2C_2O_4$ is

a. 13.5×10^{-12}
b. 1.35×10^{-12}
c. 15×10^{-12}
d. 1.5×10^{-12}

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56. Oxidation number of carbon in CO_3^{2-}, of sulphur in SO_4^{2-} and that of phosphorus in PO_4^{3-} are respectively

a. +5, +4, +6
b. +4, +5, +6
c. +4, +6, +5
d. +6, +5, +4

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57. Which of the following is not an acidic oxide?

a. CO_2
b. SiO_2
c. GeO_2
d. SnO_2

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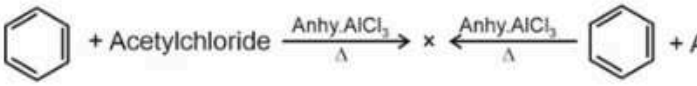
58. How many sp^2 hybridised carbon atoms and π bonds are present in the following compound?



a. 8, 5
b. 8, 6
c. 7, 5
d. 7, 6

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59. The product 'x' in the following reactions is



a. Benzoic acid
b. Acetophenone
c. Toluene
d. Benzoyl chloride

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60. Correct increasing order of bond order is

a. $O_2^+ < O_2 < O_2^- < O_2^{2-}$
b. $O_2^{2-} < O_2 < O_2^+ < O_2^-$
c. $O_2^+ < O_2^- < O_2 < O_2^{2-}$
d. $O_2^{2-} < O_2^- < O_2 < O_2^+$

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61. If 10 g of a non-electrolyte solute is dissolved in 57 g of n-octane to reduce its vapour pressure to 80%. The molar mass (in g mol^{-1}) of the solute is: (molar mass of n-octane is 114 g mol^{-1})

a. 100
b. 80
c. 90
d. 70

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62. A first order reaction has a rate constant 1.15 × 10^{-3} s^{-1}. Time required for 5 g of this reactant to reduce to 3 g is nearly

a. 444 sec
b. 44.4 sec
c. 4.44 sec
d. 221.9 sec

Pdf By: @Loop2x

63. Standard electrode potential for the Daniel cell with cell reaction:

$A(s) + B^{2+}(aq) \rightarrow A^{2+}(aq) + B(s)$ is 2.2 V. The standard Gibbs energy change for the cell reaction is: (given F = 96500 $cmol^{-1}$)

a. – 424 kJ mol^{-1}
b. + 424 kJ mol^{-1}
c. – 424 J mol^{-1}
d. + 424 J mol^{-1}

Pdf By: @Romlex69

64. Zinc reacts with dilute nitric acid to give gas (X) and with concentrated acid to give gas (Y). X and Y are

a. X = NO_2 ; Y = N_2O
b. X = N_2O ; Y = NO
c. X = NO_2 ; Y = NO
d. X = N_2O ; Y = NO_2

Pdf By: @Loop2x

65. When acidic $KMnO_4$ is treated with potassium iodide, iodide ion is converted into

a. IO^-
b. I_2
c. IO_4^-
d. IO_3^-

Pdf By: @Romlex69

66. When cobalt (III) chloride – Ammonia complex react with silver nitrate solution (in cold) to form 2 moles of silver chloride, the complex is

a. $COCl_3 \cdot 6NH_3$
b. $COCl_3 \cdot 5NH_3$
c. $COCl_3 \cdot 4NH_3$
d. $COCl_3 \cdot 3NH_3$

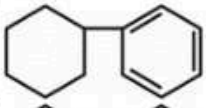
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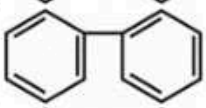
67. When chlorobenzene is treated with sodium in presence of dry ether it gives

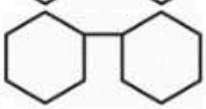
68. Which among the following has maximum number

presence of dry ether it gives

a. 

b. 

c. 

d. 

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of molecules?

a. 4.4 g CO_2
b. 4.8 g O_3
c. 4 g H_2
d. 32 g SO_2

Pdf By: @Loop2x

69. Given below are two statements.

Statement I : The radius of 2nd orbit of He atom is 1.058 Å.

Statement II : Splitting of spectral line in electrical field is called stark effect.

Choose the correct option from the following:

a. Statement I is correct but statement II is incorrect
b. Statement I is incorrect but statement II is correct
c. Both statements are incorrect
d. Both statements are correct

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70. The IUPAC name of element with atomic number 104 is

a. Unnilennium
b. Unnilquadium
c. Ununoctium
d. Unnilseptium

Pdf By: @Loop2x

71. Odd electron species among the following is

a. LiCl
b. BeF_2
c. NO_2
d. SCl_2

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72. If the bond energies of H–H, Br–Br and H–Br are a, b and c kJ mol^{-1} respectively then ΔH° for the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$ in kJ mol^{-1} will be

a. a + b –2c
b. 2c – a – b
c. c – 2a – 2b
d. a + b – 4c

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73. The compound that is not a Lewis acid is

a. $B(OH)_3$
b. $AlCl_3$
c. B_2H_6
d. NH_3

Pdf By: @Romlex69

74. If oxidation number of X, Y, Z in a compound are –3, +5 and –1 respectively then possible formula of compound will be

a. $X_2(YZ)_2$
b. $X_3(YZ_2)_2$
c. $X(Y_2Z)_2$
d. $X(YZ_4)_3$

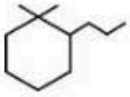
Pdf By: @Loop2x

75. In Al_2Cl_6, the bond angle of Al – Cl – Al bond is

a. 79°
b. 90°
c. 101°
d. 118°

Pdf By: @Romlex69

76. IUPAC name of the given compound is



a. 2,2-Dimethyl-1-propylcyclohexane
b. 1,1-Dimethyl-2-propylcyclohexane
c. 1,1-Dimethyl-6-propylcyclohexane
d. 6,6-Dimethyl-1-propylcyclohexane

Pdf By: @Loop2x

77. van't Hoff factor i for acetic acid in water and benzene is respectively

a. Greater than unity and greater than unity
b. Less than unity and greater than unity
c. Greater than unity and less than unity
d. Less than unity and less than unity

Pdf By: @Romlex69

78. The standard reduction potential of Mg, Al, Zn and Na are respectively –2.36 V, –1.66 V, –0.76 V and –2.71 V. The correct decreasing order of reducing power of metal is

a. Mg > Na > Al > Zn
b. Zn > Mg > Na > Al
c. Na > Mg > Al > Zn
d. Zn > Al > Mg > Na

Pdf By: @Loop2x

79. Decomposition of N_2O_5 follows first order kinetics. If concentration N_2O_5 droped from 1.24 M to 0.2 M in 60 min then rate constant will be

a. 0.304 min^{-1}
b. 0.0034 min^{-1}
c. 0.0304 min^{-1}
d. 3.04 min^{-1}

Pdf By: @Romlex69

80. In which of the following arrangements the given sequence is not strictly according to the property indicated against it?

a. HI < HBr < HCl< HF (pK_a value)
b. $H_3PO_3 < H_3PO_2 < H_4P_2O_7 < H_3PO_4$ (Number of P – OH bond(s))
c. $HClO_4 < HClO_3 < HClO_2 < HClO$ (Oxidising property)
d. $PH_3 < AsH_3 < NH_3 < SbH_3 < BiH_3$ (Boiling point)

Pdf By: @Loop2x

81. When butanal reacts with Tollen's reagent it forms

a. Butanone
b. Butanol
c. Butane
d. Butanoic acid

Pdf By: @Romlex69

82. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give :

a. Tert. butyl alcohol
b. Isobutyl alcohol
c. Isopropyl alcohol
d. Sec. butyl alcohol

Pdf By: @Loop2x

83. Primary valency of nickel in the complex $[Ni(CO)_4]$ is

a. Zero
b. 4
c. 2
d. 3

Pdf By: @Romlex69

84. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : $(CH_3)_3C$ – Cl undergoes hydrolysis reaction by S_N1 mechanism.

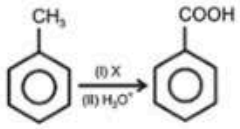
Reason (R) : $(CH_3)_3\overset{\oplus}{C}$ is very stable carbonium ion.

In the light of above statements, select the **correct**

option.

a. Both Assertion & Reason are true and the reason is the correct explanation of the assertion
b. Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
c. Assertion is true statement but Reason is false
d. Both Assertion and Reason are false statements

Pdf By: @Loop2x

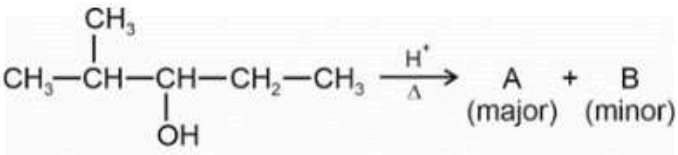
85. 

X may be

a. Conc. HNO_3
b. $KMnO_4/OH^-$
c. CrO_2Cl_2/CCl_4
d. O_3/CCl_4

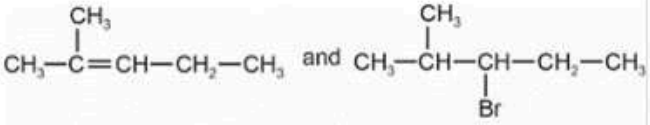
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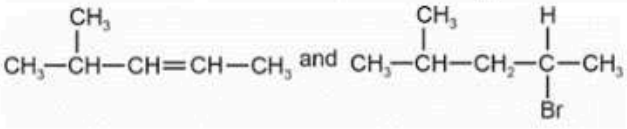
86. In the following reactions

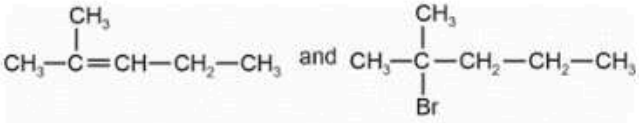


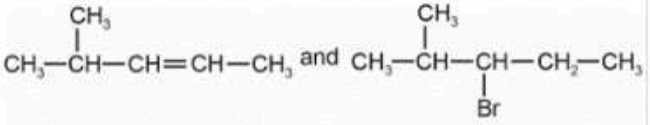


The major products A and C respectively are

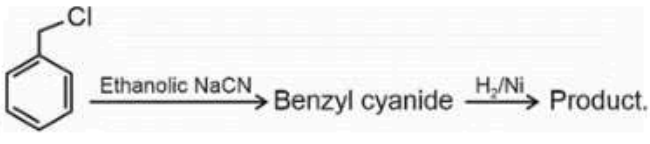
a. 

b. 

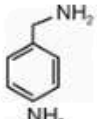
c. 

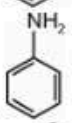
d. 

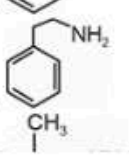
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87. 

In the above reaction the final product formed is

a. 

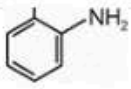
b. 

c. 

88. For titration of weak acid and strong base the suitable indicator is

a. Methyl orange
b. Methyl red
c. Phenolphthalein
d. Litmus

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d. 

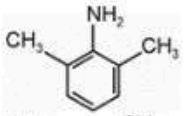
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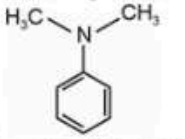
89. Glucose on reaction with mild oxidizing agent like bromine water, gets oxidized to form

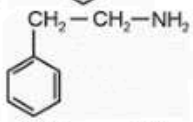
a. Glyceraldehyde
b. Glutaric acid
c. Gluconic acid
d. Saccharic acid

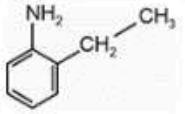
Pdf By: @Romlex69

90. The structure of N, N-Dimethylbenzenamine is

a. 

b. 

c. 

d. 

Pdf By: @Loop2x

91. In titration of Mohr's salt against $KMnO_4$, iron is oxidized. The oxidation number of iron changes from

a. +1 to +3
b. +2 to +3
c. +3 to +4
d. +1 to +2

Pdf By: @Romlex69

92. Sodium salt of benzoic acid on heating with soda lime gives

a. Benzoic acid
b. Benzene
c. Toluene
d. Phenol

Pdf By: @Loop2x

93. The mole fraction of the solute in a 4.5 m aqueous solution is approximately

a. $\frac{7}{48}$
b. $\frac{3}{40}$
c. $\frac{6}{29}$
d. $\frac{4}{58}$

Pdf By: @Romlex69

94. If the bond dissociation energies of XY, X_2 and Y_2 (all diatomic molecules) are in the ratio of 1:1:0.5 and $\Delta_f H$ for the formation of XY is –200 kJ mol^{-1}. The bond dissociation energy of X_2 will be

a. 200 kJ mol^{-1}
b. 100 kJ mol^{-1}
c. 800 kJ mol^{-1}
d. 300 kJ mol^{-1}

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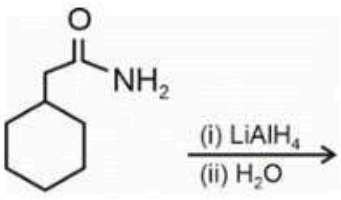
95. The electrode potentials for the following half cell reactions are

$Mg(s) \rightarrow Mg^{2+}(aq) + 2e;\ E° = +2.36\ V$

$Ag(s) \rightarrow Ag^{+}(aq) + e;\ E° = -0.80\ V$

The E^{o} for the cell reaction

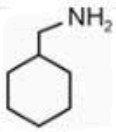
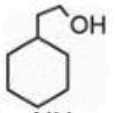
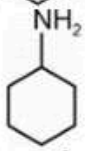
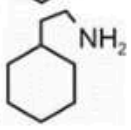
96. The product in the following reaction is



$Mg(s) + 2Ag^+(aq) \rightarrow Mg^{2+}(aq) + 2Ag(s)$, is

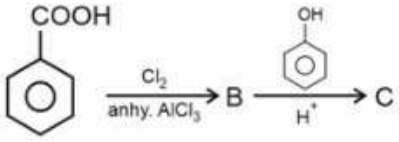
a. 3.16 V
b. 1.56 V
c. –1.56 V
d. –3.16 V

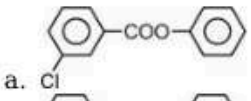
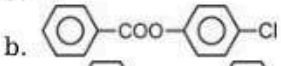
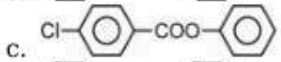
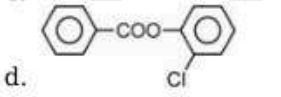
Pdf By: @Romlex69

a. 
b. 
c. 
d. 

Pdf By: @Loop2x

97. In the given sequence of reactions C is



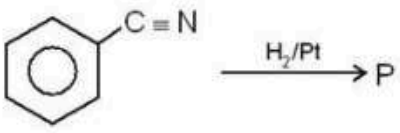
a. 
b. 
c. 
d. 

Pdf By: @Romlex69

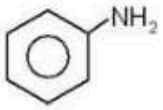
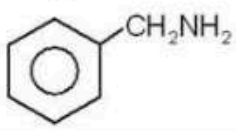
98. Electrophile in Kolbe's reaction is

a. $:CCl_2$
b. CO_2
c. $H-\overset{+}{C}=O$
d. $CH_3-\overset{+}{C}=O$

Pdf By: @Loop2x

99. 

Product P is

a. 
b. 
c. 
d. 

100. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In lanthanoids, there is a regular decrease in atomic radii with increase in atomic number of metals.

Reason (R) : In lanthanoids, the filling of 4f orbitals provides poor shielding

In the light of above statements select the correct option.

a. Both Assertion & Reason are true and the reason is the correct explanation of the assertion
b. Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
c. Assertion is true statement but Reason is false
d. Both Assertion and Reason are false statements

Pdf By: @Loop2x

Pdf By: @Romlex69

101. Read the following statements and choose the **correct** option.

Assertion: The content of nucleolus is continuous with the rest of the nucleoplasm.

Reason: Nucleolus is not a membrane bound structure.

a. Both Assertion and Reason are true and Reason is the correct explanation for Assertion
b. Both Assertion and Reason are true but Reason is not the correct explanation for Assertion
c. Only Assertion is true
d. Only Reason is true

Pdf By: @Romlex69

102. If a hybrid DNA is allowed to replicate for two generations in N^{14} containing medium then the proportion of light, hybrid and heavy DNA will be respectively

a. 75%, 25% and 0%
b. 25%, 25% and 50%
c. 0%, 50% and 50%
d. 25%, 0% and 75%

Pdf By: @Loop2x

103. Given below is the base sequences of non-coding strand of a transcription unit of DNA



How many amino acids would be coded by the m-RNA formed by this given transcription unit?

a. 4
b. 5
c. 2
d. 3

Pdf By: @Romlex69

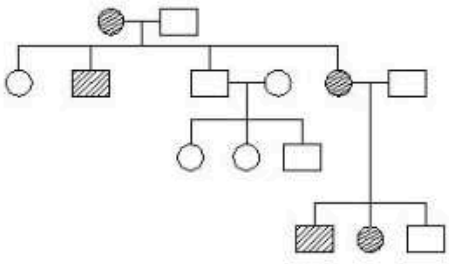
104. A heterozygous round and yellow seeded pea plant was selfed and total 1600 seeds were obtained.

Find out the number of plants develop from these seeds, that are heterozygous for seed shape but homozygous recessive for seed colour.

a. 400
b. 200
c. 800
d. 1200

Pdf By: @Loop2x

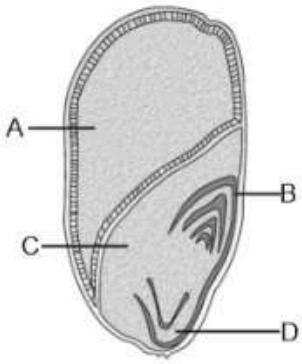
105. Given pedigree chart does **not** show the inheritance of trait like



a. Autosomal recessive trait
b. X-linked recessive trait
c. Autosomal dominant trait
d. X-linked dominant trait

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106. Identify the given figure and select the option that has all the four parts (A, B, C & D) **correctly** labelled.



	(A)	(B)	(C)	(D)
(1)	Aleurone Layer	Endosperm	Coleoptile	Coleorhiza
(2)	Endosperm	Coleoptile	Scutellum	Coleorhiza
(3)	Endosperm	Seed coat	Scutellum	Coleoptile
(4)	Seed coat	Endosperm	Coleorhiza	Coleoptile

a. (1)
b. (2)
c. (3)

c. (3)
d. (4)

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107. Read the following statements and mark the **correct** option.
(i) Grana lamellae have both PS-I and PS-II.
(ii) pH inside the thylakoid lumen decreases when H^+ gradient is established across thylakoid membrane.
(iii) During reduction of NADP to NADPH + H^+, protons are removed from thylakoid lumen.
(iv) Melvin Calvin used radioactive ^{14}C in a dicot plant to elucidate path of carbon in biosynthetic phase of photosynthesis.

a. (i) and (iii) are correct
b. (i) and (ii) are correct
c. (i) and (iv) are correct
d. (i) and (ii) are incorrect

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108. Read the following characteristics and identify the class of algae.
a. Chloroplast may be spiral, cup-shaped, discoid or reticulate.
b. Pyrenoids contain protein besides starch.
c. Some members may store food in the form of oil droplets.

a. Phaeophyceae
b. Chlorophyceae
c. Rhodophyceae
d. Cyanophyceae

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109. Tropics show rich biodiversity because

(a) Tropical areas receive more solar energy

(b) Tropical environment is relatively more constant and predictable than temperate ones.

(c) Tropical environment is less seasonal than temperate environment.

a. Only (a) is correct
b. Only (b) is correct
c. Only (c) is correct
d. All (a), (b) and (c) are correct

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110. Humus is

a. Non-decomposed living matter
b. Lighter in colour
c. Colloidal in nature and serves as a reservoir of nutrients
d. Not decomposed in moist environment

Pdf By: @Loop2x

111. Which adaptation is **not** seen in parasites?

a. Loss of unnecessary sense organs
b. Presence of adhesive organs
c. Low reproductive capacity
d. Loss of digestive system

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112. The growth status of a population is considered to be declining when

a. The individuals in post-reproductive group are lesser than those in reproductive group
b. The individuals in reproductive group are equal to those in pre-reproductive group
c. There are more individuals in pre-reproductive group as compared to reproductive group
d. There are fewer individuals in pre-reproductive group as compared to reproductive group

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113. Select the **incorrect** statement w.r.t. biogas.

a. Produced by the action of fungi only
b. Methane is its major component
c. It is used as fuel
d. It has highly Inflammable gas methane

114. Select the **correct** match.

(1) Alec Jeffrey – Proposed Clover-leaf model of tRNA

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(2) Watson and Crick – Proposed genetic code

(3) Jacob and Monod – Proposed Operon model in bacteria

(4) Nirenberg – Proposed Double helical structure of DNA

a. (1)
b. (2)
c. (3)
d. (4)

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115. Structural genes in the *lac* operon of *E. coli* get switched on when

a. Repressor binds to the operator
b. Lactose binds to the RNA polymerase
c. Lactose binds to the repressor protein
d. Inducer binds to the operator

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116. Which of the following character(s) is/are not related to Down's syndrome?

a. Short stature and small round head
b. Furrowed tongue and partially open mouth
c. Big and wrinkled tongue
d. Gynaecomastia

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117. In phenylketonuria, the amino acid *(i)* is not converted into *(ii)* . Select the **correct** option to fill in the blanks (i) & (ii).

a. (i) Glycine, (ii) Urea
b. (i) Phenylalanine, (ii) Urea
c. (i) Valine, (ii) Ketone
d. (i) Phenylalanine, (ii) Tyrosine

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118. A typical angiospermic anther is

(a) Bilobed

(b) Tetragonal

(c) Tetrasporangiate

The **correct** ones is/are

a. Only (a)
b. All (a), (b), (c)
c. Only (a) and (c)
d. Only (a) and (b)

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119. Which hormone helps to promote nutrient mobilisation and delays leaf senescence?

a. Cytokinin
b. Ethylene
c. Napthalene acetic acid
d. Abscisic acid

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120. All of the following codons have their associated t-RNAs, **except**

a. UUU
b. CAU
c. GCU
d. UAA

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121. In Krebs cycle, which of the following compounds contain maximum number of carbon atoms?

122. Plants with Kranz anatomy do **not** have/show

a. Oxaloacetic acid
b. Succinic acid
c. α-ketoglutaric acid
d. Citric acid

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a. Double fixation of CO_2
b. Calvin cycle
c. 4C acids in their mesophyll cells
d. Oxygenase activity of RuBisCO

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123. At low light intensities, increase of CO_2 concentration will

a. Increase the rate of photosynthesis in C_3 plants only
b. Increase the rate of photosynthesis in C_4 plants only
c. Increase the rate of photosynthesis in both C_3 and C_4 plants
d. Neither affect rate of photosynthesis in C_3 plants nor in C_4 plants

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124. Which of the following pair of organisms, stores food in the form of a compound that is very similar in structure to amylopectin and glycogen?

a. *Polysiphonia* and *Porphyra*
b. *Ectocarpus* and *Dictyota*
c. *Chara* and *Volvox*
d. *Fucus* and *Sargassum*

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125. Which of the following features is **not** associated with *Cycas*?

a. Presence of coralloid roots
b. Branched stem
c. Persistence of pinnate leaves for a few years
d. Presence of male cones and megasporophylls on different trees

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126. Secondary xylem and secondary phloem in dicot stem are produced by

a. Apical meristems
b. Cork cambium
c. Vascular cambium
d. Phellogen

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127. When xylem and phloem within a vascular bundle are arranged in an alternate manner along the different radii, then it is called

a. Conjoint vascular bundle
b. Closed vascular bundle
c. Radial vascular bundle
d. Open vascular bundle

Pdf By: @Romlex69

128. The floral formula of Poaceae family is

a. $Br \oplus \male \overset{\frown}{P_{(3+3)}} A_{3+3} \underline{G}_{(3)}$
b. $\% \female K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_{1}$
c. $\oplus \male K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$
d. $\% \male P_{0 \text{ or } 2 \text{ or } 3 \text{ (Lodicules)}} A_{3 \text{ or } 6} \underline{G}_{1}$

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129. The outer layer of seed coat present in gram seed is

a. Plumule
b. Testa
c. Coleoptile
d. Tegmen

130. The causative agent of potato spindle tuber disease is

a. Virus
b. Prion
c. Viroid
d. Lichen

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131. Which of the following taxonomical aids contain information on any one taxon?

a. Flora
b. Monograph
c. Manual
d. Catalogue

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132. Plant families like convolvulaceae, solanaceae are included in the order

a. Sapindales
b. Polymoniales
c. Poales
d. Diptera

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133. Exchange of genetic material between two homologous chromosomes occurs during

a. Zygotene
b. Pachytene
c. Diplotene
d. Leptotene

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134. In animal cell, centriole duplication occurs during

a. G_1 phase
b. S phase
c. G_2 phase
d. G_0 phase

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135. Which of the following cell organelle serves as the site of formation of glycoproteins and glycolipids?

a. ER
b. Vacuole
c. Lysosome
d. Golgi bodies

Pdf By: @Romlex69

136. In logistic growth, a plot of population density (N) in relation to time (t) results in a sigmoid curve, which finally shows an asymptote when

a. Intrinsic rate of natural increase is one
b. Carrying capacity = Population density
c. Carrying capacity > Population density
d. Carrying capacity < Population density

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137. Match the columns and select the **correct** option.

Column I	Column II
a. *Ustilago*	(i) Zygospores
b. *Claviceps*	(ii) Basidiospores
c. *Mucor*	(iii) Conidia
d. *Trichoderma*	(iv) Ascospores

a. a(ii), b(i), c(iv), d(iii)
b. a(iii), b(iv), c(i), d(ii)
c. a(ii), b(iii), c(iv), d(i)
d. a(ii), b(iv), c(i), d(iii)

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138. Select the **odd** one w.r.t. species-area relationship curve proposed by Alexander von Humboldt

a. The relation between species evenness and area for a wide variety of taxa turns out to be a rectangular hyperbola
b. Z value for plants in Britain lies between 0.1 to 0.2
c. Regression coefficient for frugivorous birds and mammals in the tropical forests of different continents is 1.15
d. Slope of the line will become steeper for species-area relationships among very large areas like entire continent

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139. Which of the following option exemplify an inverted pyramid?

a. Pyramid of energy in aquatic ecosystem
b. Pyramid of biomass in sea

140. BOD measures

a. Rate of oxygen release by micro-organisms in a sample of water
b. Amount of inorganic matter present in the water

c. Pyramid of number in grassland ecosystem
d. Pyramid of energy in forest ecosystem

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c. Polluting potential of water
d. Enhancement in the dissolved oxygen levels of only pure water

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141. In eukaryotes, RNA polymerase III transcribes

a. hnRNA
b. 28S rRNA
c. 5.8S rRNA
d. snRNA

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142. Female heterogamety can be observed in

a. *Drosophila*
b. Human
c. Birds
d. Grasshopper

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143. Sporopollenin is found as a constituent in

a. Intine
b. Exine
c. Germ pore
d. Generative cell

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144. Which of the following plants exhibit heterophyllous development due to environment?

a. Buttercup
b. Cotton
c. Coriander
d. Larkspur

Pdf By: @Loop2x

145. Invertase enzyme catalyzes the conversion of

a. Sucrose into glucose and fructose
b. 1, 3 bisphosphoglyceric acid into 3-phosphoglyceric acid
c. Phosphoenol pyruvate into pyruvic acid
d. Glyceraldehyde-3-phosphate into DHAP

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146. Which of the following is **not** a product of photochemical reaction of photosynthesis?

a. CO_2
b. O_2
c. ATP
d. NADPH

Pdf By: @Loop2x

147. Which of the following alga shows diplontic life cycle?

a. *Fucus*
b. *Ectocarpus*
c. *Polysiphonia*
d. *Chlamydomonas*

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148. Starch sheath can be found in

a. Dicot stem
b. Dicot leaf
c. Monocot leaf
d. Monocot root

Pdf By: @Loop2x

149. All of the following plants have syncarpous (gynoecium), **except**

a. Lotus
b. Mustard
c. Tomato
d. *Petunia*

Pdf By: @Romlex69

150. In which of the following chromosome, terminal centromere can be observed?

a. Metacentric chromosome
b. Sub-metacentric chromosome
c. Telocentric chromosome
d. Acrocentric chromosome

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151. Which one of the following groups of three animals is correctly matched with their characteristic feature?

	Organisms		Characteristic Feature
(1)	*Aptenodytes, Adamsia, Meandrina*	–	Bioluminescence
(2)	*Anopheles, Ascaris, Ancylostoma*	–	Presence of Pseudocoelom
(3)	*Locusta, Ascaris, Laccifer*	–	Jointed appendages
(4)	*Echinus, Asterias, Antedon*	–	Water vascular system

a. (1)
b. (2)
c. (3)
d. (4)

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152. Select the odd one from the options given below w.r.t. organisms in which notochord is persistent throughout the life.

a. Dog fish
b. Saw fish
c. Great white shark
d. Angel fish

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153. Read the features listed below.

(a) Fibres and fibroblasts are compactly packed

(b) Collagen fibres are present in rows between many parallel bundles of fibres

(c) Present in structure which connects skeletal muscles to bones

Select the correct option that represents the above mentioned features.

a. Neural tissue
b. Areolar tissue
c. Dense regular connective tissues
d. Dense irregular connective tissue

Pdf By: @Romlex69

154. Biologists describe structure of which of the following biomolecules at four levels?

a. Nucleotide
b. Protein
c. Carbohydrates
d. Lipids

Pdf By: @Loop2x

155. Number of nitrogen atoms in a component of nucleotide that pairs with nitrogen base found in DNA but not in RNA, is

a. 4
b. 5
c. 7
d. 9

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156. Select the option that correctly represents the conditions occurring during the inspiration in humans under normal physiological conditions.

	External Inter-costal muscle	Diaphragm	Volume of thoracic cavity	Intrapulmonary pressure
(1)	Contraction	Contraction	Decreases	Decreases
(2)	Contraction	Relaxation	Increases	Decreases
(3)	Contraction	Contraction	Increases	Decreases
(4)	Relaxation	Contraction	Decreases	Increases

a. (1)
b. (2)
c. (3)
d. (4)

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157. The pO_2 in systemic veins is

a. More than the pCO_2 of systemic arteries
b. Less than the pCO_2 in alveolus
c. Equal to the pO_2 in pulmonary vein
d. Less than the pO_2 of systemic arteries

Pdf By: @Romlex69

158. A patient 'X' has nil antibodies in its blood plasma against blood group antigens. Select the option that correctly depicts its donor group(s).

a. A and B only
b. A and O only
c. O only
d. AB, A, B and O

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159. Select the incorrect statement w.r.t. humans.

a. Cardiac output can be defined as the volume of blood pumped out by each ventricle per minute.
b. The first heart sound is associated with the closure of the semilunar valves.
c. By counting the number of QRS complexes that occurs in a given time period, one can determine the heart rate of an individual.
d. A unique vascular connection that exists between the digestive tract and liver is called hepatic portal system.

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160. A fall in glomerular blood pressure can activate the __'A'__ cells to release __'B'__ which converts angiotensinogen in blood to __'C'__ .
Select the correct option to fill in the blanks for A, B and C.

	A	B	C
(1)	J.G.	Renin	Angiotensin I
(2)	Renin	Aldosterone	Angiotensin II
(3)	J.G.	Angiotensin I	Renin
(4)	J.G.	Angiotensin II	Angiotensin I

a. (1)
b. (2)
c. (3)
d. (4)

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161. Select the organism that excretes a nitrogenous waste that is neither most toxic form nor least toxic and a small amount of this waste is retained in the medullary interstitium of kidneys of these animals to maintain osmolarity.

a. *Clarias*
b. *Calotes*
c. *Homo*
d. *Columba*

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162. White muscle fibres differ from red muscle fibres in

a. Having high content of myoglobin
b. Containing plenty of mitochondria
c. Having reddish appearance
d. Having high amount of sarcoplasmic reticulum

Pdf By: @Loop2x

163. Age-related disorder, characterised by decreased bone mass and increased chances of fractures of bones is

a. Myasthenia gravis
b. Tetany
c. Muscular dystrophy
d. Osteoporosis

Pdf By: @Romlex69

164. In humans, 2 systems jointly coordinate and regulate the physiological functions in the body. One of them is responsible for providing organised network of point to point connection.

Select the odd one w.r.t. the secretion of other system involved.

a. Adrenaline
b. Thyroxine
c. Mucus
d. Somatostatin

Pdf By: @Loop2x

165. All of the following diseases are caused due to hyposecretion of hormones, except

a. Dwarfism
b. Diabetes mellitus
c. Cretinism
d. Acromegaly

Pdf By: @Romlex69

166. Two hormones 'A' and 'B' interact with intracellular receptors and regulate the gene expression. 'A' is steroidal and 'B' is non steroidal in nature. Select the option that correctly identifies 'A' and 'B' respectively.

a. Aldosterone, Progesterone
b. Progesterone, Growth hormone
c. Testosterone, Thyroxine
d. Thyroxine, Estrogen

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167. Match column I with column II w.r.t embryonic development in humans and select the option with all the correct matches.

	Column I		**Column II**
(i)	Embryo heart is formed	(a)	End of 24th weeks of pregnancy
(ii)	Foetus develops limbs and digits	(b)	After one month of pregnancy
(iii)	Limbs and external genital organs are well developed	(c)	End of 2nd month of gestation
(iv)	The body is covered with fine hairs, eyelids separate and eyelashes are formed	(d)	End of 12 weeks of gestation

a. (i)-b, (ii)-c, (iii)-d, (iv)-a
b. (i)-a, (ii)-c, (iii)-b, (iv)-d
c. (i)-d, (ii)-a, (iii)-b, (iv)-c
d. (i)-b, (ii)-d, (iii)-c, (iv)-a

Pdf By: @Romlex69

168. While performing gel electrophoresis, DNA fragments move in response to an applied electric field. The band of DNA fragments farthest from the loading site represents the

a. Shortest fragment of DNA
b. Longest fragment of DNA
c. Moderate sized fragments of DNA
d. Joining of DNA fragments

Pdf By: @Loop2x

169. Read the following statements

Statement I In amniocentesis some of the amniotic fluid of the developing foetus is taken to analyze the fetal cells.

Statement II This procedure is used to test the presence of metabolic disorders like typhoid, cancer, rabies, *etc.*

Choose the correct option

a. Both statement I and statement II are true correct.
b. Both statement I and statement II are false.
c. Statement I is true, statement II is false
d. Statement I is false, statement II is true

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170. Match column I with column II w.r.t. contraceptives and their mode of action.

	Column I		**Column II**
(i)	Vaults	(a)	Inhibit ovulation
(ii)	Multiload 375	(b)	Blocks gamete transport
(iii)	Vasectomy	(c)	Suppresses sperm motility
(iv)	Steroidal oral pills	(d)	Block the entry of sperms through the cervix

Choose the option with all correct matches.

a. (i)-c, (ii)-b, (iii)-a, (iv)-d
b. (i)-d, (ii)-c, (iii)-b, (iv)-a
c. (i)-a, (ii)-c, (iii)-b, (iv)-d
d. (i)-d, (ii)-c, (iii)-a, (iv)-b

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171. Select the correct match w.r.t. hominids, their cranial capacities and characteristics.

	Hominid	Cranial capacity	Characteristics
(1)	*Homo habilis*	650–800 cc	First human-like being the hominid
(2)	*Homo erectus*	900 cc	Was more ape like
(3)	*Homo sapiens*	1100 cc	They were probably not taller than 4 feet
(4)	Neanderthal man	700 cc	They used hides to protect their body

a. (1)
b. (2)
c. (3)
d. (4)

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172. Choose the incorrect match w.r.t. life cycle of *Plasmodium*.

a. Infective stage for humans → Sporozoites
b. Fertilization site → Mosquito's gut
c. Formation of sporozoites → Liver cells of human
d. Formation of gametocytes → RBCs of human

Pdf By: @Loop2x

173. Select the incorrect statement.

a. *Cry*1Ac and *cry*1Ab control corn borer and *cry*1Ab controls cotton bollworms.
b. During RNA interference, the source of complementary RNA could be from an infection by virus having RNA genomes or transposons.
c. Insulin is synthesised in our body as prohormone which contains an extra stretch called the c-peptide.
d. Recombinant DNA Technology, Polymerase Chain Reaction and Enzyme Linked Immuno-sorbent Assay are some of the techniques that serve the purpose of early diagnosis.

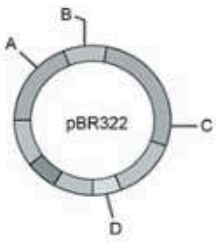
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174. Select the odd one w.r.t. forebrain of frog.

a. Olfactory lobes
b. Paired cerebral hemispheres
c. Unpaired diencephalon
d. Optic lobes

Pdf By: @Loop2x

175. In following plasmid pBR322, insertion of a foreign DNA at which of the restriction site will inactivate the ampicillin resistance gene?



a. A
b. B
c. C
d. D

Pdf By: @Romlex69

176. Nematode specific genes were introduced into the tobacco plant by using

a. pBR 322
b. Ti plasmid
c. Retro virus
d. Bacteriophage

Pdf By: @Loop2x

177. If a foreign gene is ligated at *Pst* I site in pBR322;

178. Which of the following is not obtained from the

which of the following statements is **correct**?

a. Recombinants will grow in ampicillin containing medium
b. Recombinants will grow in both ampicillin and tetracycline containing mediums
c. Recombinants will not grow in medium lacking antibiotics
d. Recombinants will grow in tetracycline containing medium

Pdf By: @Romlex69

plant *Cannabis sativa*?

a. Marijuana
b. Hashish
c. Charas
d. Heroine

Pdf By: @Loop2x

179. An adult human male ejaculates 200 - 300 million sperms per ejaculation. For normal fertility, at least what per cent of sperms should have normal shape, size and vigorous motility?

a. 60
b. 24
c. 20
d. 50

Pdf By: @Romlex69

180. Choose the **incorrect** statement regarding the epithelial tissue.

a. There are specialised junctions which provide structural and functional links between cells.
b. Protection is the main function of compound epithelium.
c. There is large amount of intercellular material between the cells.
d. The tissue has a free surface which faces either a body fluid or the outside environment.

Pdf By: @Loop2x

181. Which amongst the following structures is absent in the body of adult *Rana tigrina*?

a. Tympanum
b. Nictitating membrane
c. Sinus venosus
d. Pinna

Pdf By: @Romlex69

182. Glial cells in humans are

a. Excitable cells present in muscles
b. Structural and functional units of neural tissue
c. Make up less than one-third the volume of neural tissue in our body
d. Non-excitable cells of neural tissue

Pdf By: @Loop2x

183. The operation of natural selection on different traits shows two peaks formation in

a. Disruptive selection
b. Directional selection
c. Stabilising selection
d. Balancing selection

Pdf By: @Romlex69

184. In rDNA technology, during the process of isolation of the genetic material, purified DNA is ultimately precipitated out after the addition of

a. Ice
b. Calcium
c. Ethidium bromide
d. Chilled ethanol

Pdf By: @Loop2x

185. Decline in which of the following conditions, will **not** lead to increase in the rate of growth of population?

a. Death rate
b. Infant mortality rate
c. Maternal mortality rate
d. Number of people in reproducible age

Pdf By: @Romlex69

186. How many of the factors given in the box below will affect the activity of enzymes?

Temperature, pH, Substrate concentration, Enzyme concentration, Activation energy

a. Five
b. Four
c. Three
d. Two

Pdf By: @Loop2x

187. Which of the following set of ions/molecules are generally secreted in PCT of nephron?

a. H^+ and Na^+
b. Na^+ and HCO_3^-
c. H^+ and NH_3
d. NH_4^+ and Na^+

Pdf By: @Romlex69

188. A tract of nerve fibres that connects the left and right cerebral hemispheres is called

a. Corpus callosum
b. Association area
c. Synapse
d. Corpora quadrigemina

Pdf By: @Loop2x

189. Type of joint in humans that does **not** help in locomotion and any movements is called

a. Hinge joint
b. Gliding joint
c. Cartilaginous joint
d. Fibrous joint

Pdf By: @Romlex69

190. Which of the following structures of proteins gives an idea about the positional information of amino acids?

a. Primary structure
b. Secondary structure
c. Tertiary structure
d. Quaternary structure

Pdf By: @Loop2x

191. Which of the following animals is characterised by the presence of crop and gizzard in its digestive system?

a. *Pteropus*
b. *Aptenodytes*
c. *Balaenoptera*
d. *Delphinus*

Pdf By: @Romlex69

192. Select the **correct** order of flow of milk secreted by alveoli of mammary gland in humans.

a. Mammary tubule → Mammary alveoli → Mammary duct → Mammary ampulla → Lactiferous duct
b. Mammary alveoli → Mammary tubule → Mammary duct → Mammary ampulla → Lactiferous duct
c. Mammary alveoli → Mammary tubule → Mammary ampulla → Mammary duct → Lactiferous duct
d. Mammary ampulla → Mammary alveoli → Mammary tubule → Mammary duct → Lactiferous duct

Pdf By: @Loop2x

193. Cycads and angiosperms arose from a common ancestor called _____ that existed on earth during ____ period.

Choose the option that correctly fills the blanks

a. *Zosterophyllum*, triassic
b. Chlorophyta, silurian
c. Seed ferns, carboniferous
d. Herbaceous lycopods, cretaceous

Pdf By: @Romlex69

194. Select the option that includes only medicated IUDs.

a. Vaults, LNG-20
b. Multiload 375, Lippes loop
c. Cu7, Progestasert
d. LNG-20, Lippes loop

Pdf By: @Loop2x

195. Birds can be differentiated from mammals on the

196. The first pair of wings of cockroach arises from

195. Birds can be differentiated from mammals on the basis of presence of

a. Four chambered heart
b. Hollow bony endoskeleton with air cavities
c. Homeothermy
d. Pulmonary respiration

Pdf By: @Romlex69

196. The first pair of wings of cockroach arises from __A__ and is known as __B__.

Select the option which fills the blanks A and B **correctly**.

	A	**B**
(1)	Metathorax	Metathoracic wings
(2)	Mesothorax	Tegmina
(3)	Mesothorax	Metathoracic wings
(4)	Metathorax	Tegmina

a. (1)
b. (2)
c. (3)
d. (4)

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197. Which of the following does not involve in the systemic circulation?

a. Right atrium
b. Left ventricle
c. Pulmonary vein
d. Vena cava

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198. Match column I with column II and select the **correct** option w.r.t. antagonism.

Column I	**Column II**
a. Insulin	(i) Glucagon
b. Calcitonin	(ii) Parathormone
c. Aldosterone	(iii) Somatostatin
d. GHRH	(iv) Atrial natriuretic factor

a. a(i), b(iv), c(ii), d(iii)
b. a(i), b(iii), c(iv), d(ii)
c. a(ii), b(i), c(iv), d(iii)
d. a(i), b(ii), c(iv), d(iii)

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199. At present __A__ recombinant therapeutics being marketed in India whereas about __B___ recombinant therapeutics have been approved world over for human-use. **Correct** option for A and B is

a. **A** – 30, **B** – 12
b. **A** – 12, **B** – 30
c. **A** – 36, **B** – 24
d. **A** – 40, **B** – 30

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200. 'The integration of natural science and organisms, cells, parts thereof, and molecular analogues for products and services'.

This definition of biotechnology is given by

a. CDRI
b. GEAC
c. EFB
d. IUB

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